

INTER-IIT Tech Meet Report

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1 Problem Understanding

The primary objective is to derive elemental abundances and compositional distributions of the Moon’s surface by leveraging Chandrayaan-2’s CLASS (Chandrayaan-2 Large Area Soft X-ray Spectrometer) and XSM (Solar X-ray Monitor) datasets. This involves identifying the best-fit elemental ratios (e.g., Mg/Si, Al/Si) and creating high-resolution 3D compositional maps. Preprocessing includes removing geotail-contaminated intervals, which introduce spurious signals due to Earth’s magnetosphere, and retaining only Good Time Intervals (GTIs) with stable solar X-ray flux and minimal noise. The CLASS data, recorded at 8-second intervals, is aggregated into 32-second blocks to optimize spatial and temporal resolution, enabling precise plotting of best-fit ratios and generating detailed 3D abundance maps to capture compositional heterogeneity of the lunar surface.

2 Final Approach, outline of steps taken and algorithm

2.1 Introduction

The Chandrayaan-2 mission’s CLASS (Chandrayaan-2 Large Area Soft X-ray Spectrometer) and XSM (Solar X-ray Monitor) datasets are key to deriving elemental abundances of the lunar surface. To ensure high precision, the raw data undergoes rigorous preprocessing. This report describes the step-by-step preprocessing workflow, including geotail contamination removal, Good Time Interval (GTI) filtering, and temporal-spatial data restructuring.

2.2 Data Acquisition

The raw CLASS and XSM datasets were obtained from the **PRAGYAN Chandrayaan-2 data repository**. The CLASS data provides X-ray fluorescence (XRF) line intensities corresponding to the lunar surface, while the XSM data records the incident solar spectrum, essential for calibrating these fluorescence signals.

2.3 Preprocessing Workflow

2.3.1 Removal of Geotail Contaminated Data

Understanding Geotail Contamination in Lunar Observations

The geotail, a feature of Earth’s magnetosphere, extends away from the Earth in the direction opposite to the Sun due to solar wind interactions. It is an elongated region populated with high-energy charged particles, which can significantly impact the quality of space-based X-ray observations, including those by Chandrayaan-2’s CLASS (Chandrayaan-2 Large Area Soft X-ray Spectrometer). CLASS operates by detecting

X-ray fluorescence (XRF) from the lunar surface, which provides critical insights into the Moon's elemental composition. However, when the spacecraft traverses the geotail, these charged particles introduce spurious signals that manifest as noise or anomalies in the X-ray data, particularly in the high-energy continuum regions of the spectrum.

Importance of Geotail Removal

Data contamination from geotail crossings can lead to misinterpretation of XRF signals, particularly by introducing extraneous features such as Al- K_α and Cu- K_α peaks originating from particle interactions with the spacecraft's onboard materials. These interferences obscure the true XRF signals from the lunar surface, making preprocessing to remove geotail-contaminated intervals crucial for generating reliable elemental maps.

SPICE Kernel Data

- **Spacecraft Planet Kernel (SPK)**
 - **File Example:** `ch2_orb_DDMMYYYYY_ddmmmyyyy_vX.bsp`
 - **Purpose:** SPK files contain ephemeris data for celestial bodies and spacecraft. They store time-stamped position (and sometimes velocity) data of the spacecraft relative to other bodies, allowing for the determination of where the spacecraft was (or will be) at a given time.
- **Spacecraft Clock Kernel (SCLK)**
 - **File Example:** `ch2_sclk_vX.tsc`
 - **Purpose:** SCLK files provide the information needed to convert between the spacecraft's on-board clock time and standard Ephemeris Time (ET). Since spacecraft clocks can drift or differ from Earth-based timescales, these kernels enable accurate correlation of spacecraft event times with universal time frames.
- **Leap Seconds Kernel (LSK)**
 - **File Extension:** `.tls`
 - **Purpose:** LSK files contain the leap second records used to correlate Universal Time Coordinated (UTC) with Barycentric Dynamical Time (TDB) and ET. These kernels ensure correct conversion between Earth-based civil time (UTC) and the spacecraft's internal timing systems (ET), accounting for the irregular addition of leap seconds over time.
- **Planetary Constants Kernel (PCK)**

- **File Extension:** `.tpc`
- **Purpose:** PCK files provide physical, dynamic, and cartographic parameters of solar system bodies, such as planetary radii, orientation data, and spin characteristics. Although not always directly needed for simple position calculations, they are essential when details about the shape, rotation, or gravity of a target body are required—particularly for mapping, surface analysis, and geometric modeling.

Approach to Geotail Detection

1. Data Integration:

- SPICE ephemeris kernels were loaded to access precise spacecraft positional data and correlate it with lunar orbital parameters.
- Background noise modeling was carried out using X-ray spectral data recorded during periods without geotail interference.

2. Defining Geotail Intervals:

- Positional data of Chandrayaan-2 were analyzed to identify intervals when the spacecraft was within Earth’s magnetospheric tail, corresponding to periods around the full Moon when Earth blocks solar wind.
- Spectral anomalies, particularly elevated particle-induced counts above a threshold, were flagged as geotail crossings.

3. Background Noise:

- **Identification of High Solar Angle Files:** The code traverses through a directory structure and filters FITS files based on the `SOLARANG` header keyword. Only those files with a solar angle greater than 120 degrees are selected to represent conditions associated with higher background levels.
- **Cumulative Counts and Averaging:** For each qualifying FITS file, the code extracts `CHANNEL` and `COUNTS` columns and accumulates these counts across all selected files. By averaging the summed counts, an approximate baseline “background noise” level is derived.

4. Threshold-based Filtering:

- Energy bins exceeding background levels by more than a predefined threshold (0.7 counts) were marked as contaminated.
- Contaminated intervals were logged and removed from further analysis.

2.4 Spectral Characteristics and Challenges

During geotail crossings, spectral signatures are heavily distorted by particle-induced X-rays, such as those from the spacecraft’s aluminum filters and copper collimators. This contamination is evident as elevated counts in the 1.5 keV and 8 keV regions, which correspond to Al and Cu respectively. The variability in particle flux during geotail passage necessitates a dynamic approach to contamination detection.

2.5 Mitigation and Results

The geotail detection algorithm effectively identified intervals with significant contamination, enabling their exclusion from the dataset. This preprocessing ensures that the remaining data reflect only the lunar surface’s XRF, devoid of extraneous signals. The algorithm’s integration of SPICE data, spectral analysis, and robust noise modeling contributes to its high accuracy in isolating contaminated intervals.

This improved preprocessing workflow ensures the quality and reliability of Chandrayaan-2 CLASS data, supporting the mission’s objective of creating high-resolution, contamination-free elemental abundance maps of the Moon.

2.6 Further Data Processing

- 1. Identification of Good Time Intervals (GTIs):** Initially, raw GTI data files were accessed for each day within a specified month. Each file was checked for the presence of good intervals, ensuring that only those segments meeting quality criteria were retained. These GTIs were then converted from Mission Elapsed Time (MET) to Universal Coordinated Time (UTC) to maintain a consistent and standard time reference. These GTI files were extracted from XSM data.
- 2. Filtering and Intersection of Intervals:** Two sets of intervals were given: the established 8-second intervals and the identified GTIs. These two sets were intersected to ensure that only the segments fitting both criteria—i.e., good quality and correct duration—were selected for further processing. After sorting and intersecting, the final list of valid 8-second intervals was exported to a CSV file.
- 3. Ensuring Interval Continuity and Forming 32-Second Batches:** Since the analysis required longer durations, the next step involved grouping sets of four consecutive 8-second intervals to form continuous 32-second segments. Each candidate set was checked for continuity to ensure that the end time of one interval matched exactly the start time of the next. Only continuous and valid sets of four 8-second intervals were accepted, thereby producing discrete 32-second batches of data.

4. **Association with Corresponding FITS Files:** With the final 32-second batches defined, the process continued by attempting to match each interval batch with corresponding FITS data files. Since filenames were time-stamped, a standardized approach was used to generate expected filenames without millisecond components and then search the dataset directories for matching FITS files. Intervals that could not be matched to a FITS file were noted for further inspection.
5. **Extraction of Counts and Cumulative Analysis:** Once matching FITS files were located, the relevant data columns (e.g., `CHANNEL` and `COUNTS`) were extracted. These channel indices were converted into energy values (keV), and the counts from each FITS file contributing to a given batch were summed cumulatively. This cumulative aggregation allowed for the creation of batch-level spectra, which were then plotted and saved as both image files and CSV tables. This provided a clear visualization and record of the energy-dependent counts over the 32-second intervals.

2.7 Data Processing and Analysis

Building upon the preprocessing workflow, a comprehensive data processing and analysis pipeline was developed to accurately derive elemental abundances and compositional distributions of the Moon’s surface. This section delineates the systematic steps undertaken, from flare identification to elemental ratio calculation, ensuring the robustness and reliability of the results.

2.7.1 Flare Identification and Classification

To ensure the integrity of the dataset, it was imperative to identify and classify periods of solar flares, which can significantly influence the X-ray measurements. Utilizing the **Solar X-ray Monitor (XSM)** data’s light curve file, the total count of X-ray photons was visualized on a daily basis. High solar flare activities introduce substantial variability in the X-ray flux, necessitating their identification and classification.

Visualization of Daily Counts

The XSM light curve data was first visualized to monitor the total X-ray counts per day. This visualization facilitated the identification of anomalously high count rates indicative of solar flare events. By plotting the daily counts, periods of elevated activity became readily apparent, allowing for subsequent classification.

Classification Thresholds for Flare Levels

- **C-Level Flares:** Identified by a cutoff threshold of 1,400 counts. Events exceeding this threshold were categorized as C-level flares.
- **M-Level Flares:** Identified by a higher cutoff threshold of 14,000 counts. Events surpassing this threshold were classified as M-level flares.

2.7.2 Data Segmentation Based on Flare Activity

Post classification, the dataset was segmented into three distinct categories to streamline the analysis process:

1. **C-Level Flares**
2. **M-Level Flares**
3. **No Flare**

Each segment comprised time intervals that matched the **Good Time Intervals (GTIs)** identified from the XSM data.

2.7.3 Background Subtraction

Accurate elemental abundance determination necessitates the removal of background noise from the X-ray measurements. A background file was meticulously created using the **No Flare** segments, which represent periods devoid of significant solar flare activity. This background was then subtracted from the high flare level datasets (both C and M-level flares) to isolate the genuine XRF signals originating from the lunar surface.

2.7.4 Assumptions in Data Calibration

For the scope of this analysis, it was assumed that the provided data had already been calibrated to the **Auxiliary Response File (ARF)** standard. Consequently, no additional calibration steps were deemed necessary.

2.7.5 Aggregation of FITS Files into 32-Second Intervals

The raw **FITS** files, recorded at 8-second intervals, were aggregated into 32-second blocks. The aggregation process involved summing the counts from four consecutive 8-second FITS files to form a coherent 32-second dataset.

2.7.6 Spectral Smoothing and Peak Detection

A Gaussian smoothing algorithm with a sensitivity parameter of 2 was applied, and peaks were identified using a prominence threshold of 0.5% of the total counts. Gaussian curves were fitted to each peak to quantify elemental abundances.

2.7.7 Filtering Based on Total Counts

FITS files with a total sum of counts less than 3,000 were disregarded, ensuring only spectra with sufficient statistical significance were included.

2.7.8 Elemental Ratio Calculation

The area under each elemental peak's Gaussian curve was divided by the area under the silicon peak's Gaussian curve to produce elemental ratios ($\frac{\text{Area}_{\text{Element}}}{\text{Area}_{\text{Si}}}$).

2.8 Summary

In summary, this workflow efficiently combined time interval selection, data validation, file matching, and cumulative spectral analysis into a coherent pipeline. The result was a set of reliable 32-second integrated datasets, each thoroughly documented, easily visualized, and prepared for subsequent scientific investigation.

The geotail is an elongated region of Earth's magnetosphere characterized by high-energy particle fluxes. During geotail traversal, CLASS and XSM instruments are subject to severe contamination, making the data unsuitable for scientific analysis.

Steps Taken:

- removing data contaminated by geotail traversal, which introduces spurious signals due to particle interactions in Earth's magnetosphere.
- Geotail intervals were identified using spacecraft positional data and magnetospheric models.
- These intervals were entirely excluded from both CLASS and XSM datasets.

GTIs are intervals during which the solar X-ray flux is stable, free from solar flares, and unaffected by instrumental or particle noise. Retaining only GTI-compliant data ensures high-quality XRF signals.

Post geotail removal, data was filtered using Good Time Intervals (GTIs), identified from XSM data. GTIs represent periods of stable observational conditions.

Steps Taken:

- GTIs were determined based on stability in XSM flux measurements and particle noise analysis.
- Data outside GTIs were removed to enhance reliability.

The CLASS data was initially recorded at 8-second intervals, corresponding to a spatial footprint of 12.5×12.5 km. This temporal granularity required aggregation for certain analyses.

Conversion to 32-Second Data Blocks

The processed dataset was reorganized into 32-second time blocks by aggregating 4 consecutive 8-second intervals. This step provides a larger spatial footprint for compositional mapping while retaining sufficient resolution.

Spatial Characteristics

- Each 32-second data block corresponds to a footprint of 50 km (along-track) $\times$ 12.5 km (across-track).
- This aggregation improves statistical robustness by averaging out short-term fluctuations while preserving spatial resolution.

3 Best ratios to use and visualization of data on a lunar map

The primary elemental ratios used for compositional mapping include:

- Magnesium to Silicon (Mg/Si)
- Aluminum to Silicon (Al/Si)
- Calcium to Silicon (Ca/Si)

These ratios are visualized using a 3D lunar base map, enabling the representation of compositional heterogeneity.

Dynamic Mapping and Visualization of Lunar Data (Subsection)

Initially, we utilized QGIS for visualizing and plotting our data, which proved to be effective for static visualizations. However, we encountered significant challenges while attempting to implement dynamic mapping. The primary issue stemmed from synchronization errors between the latitude-longitude coordinates in our dataset and those of the

original dynamic lunar map. These discrepancies resulted in inaccurate spatial representations, hindering our ability to overlay and analyze data correctly.

To address these challenges, we developed a robust prototype model designed to facilitate dynamic mapping with enhanced precision. This model processes input from a CSV file, which contains critical data such as latitude, longitude, and derived outputs like elemental ratios. By integrating the NASA Moon Trek API, we leveraged its high-resolution lunar maps and precise geographic coordinates to align our dataset with the lunar surface effectively.

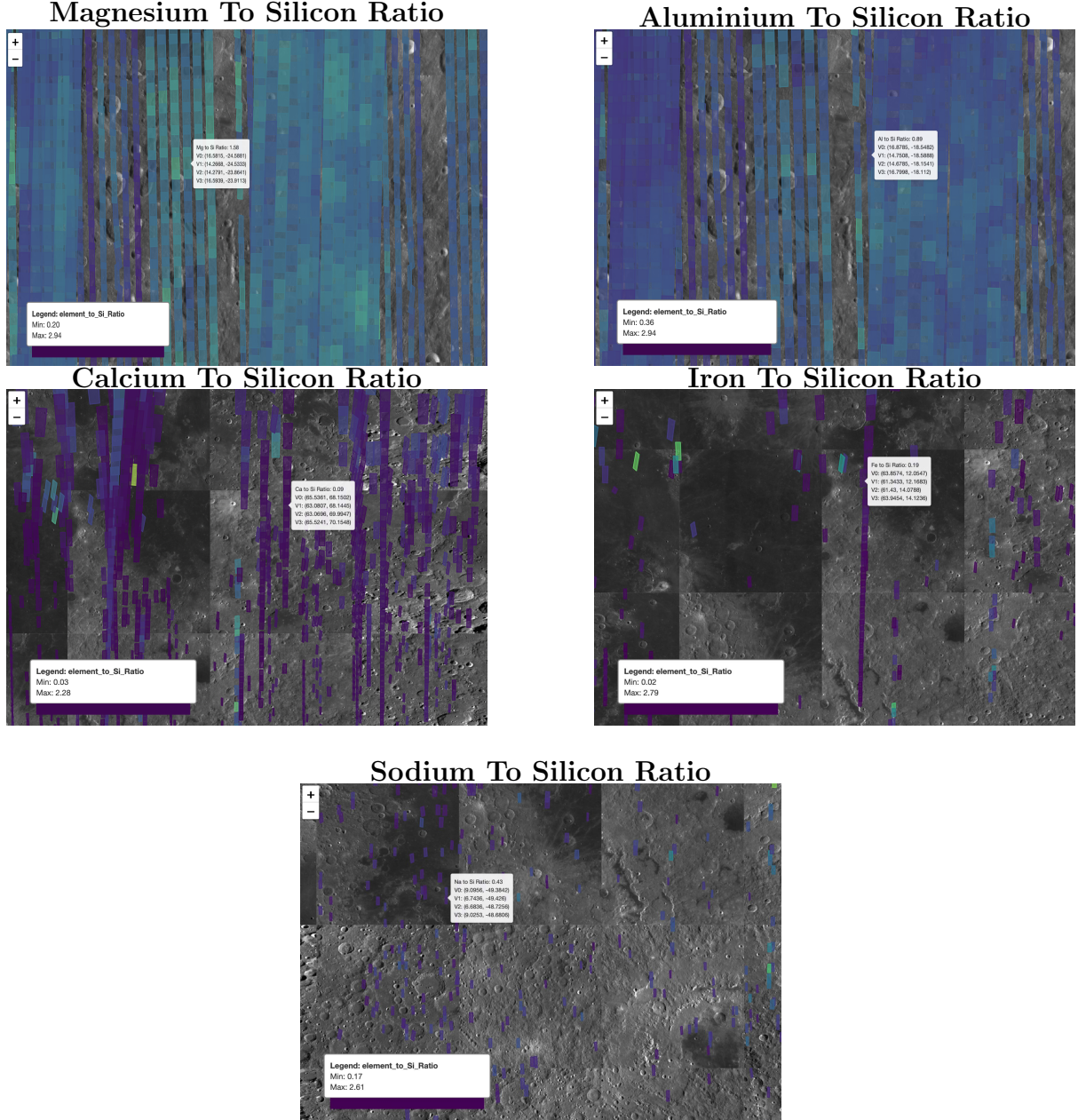
Our Python-based implementation systematically compares the latitude and longitude coordinates in our dataset, represented as rectangular coordinate areas, with those obtained from the NASA Moon Trek API. The model iteratively checks for matches between these coordinates and, upon finding a match, marks the corresponding location on the lunar map with the computed elemental ratio or other desired results. This process ensures accurate alignment and mapping of our data onto the lunar surface, enabling reliable spatial analysis.

To further improve usability, we developed a dynamic user interface using HTML. This interface allows users to interact with the mapped data in real time, providing functionalities for visualizing results dynamically and exploring insights seamlessly. The integration of a dynamic web interface enhances the accessibility and flexibility of our mapping approach, making it a comprehensive solution for lunar data visualization and analysis.

Our approach successfully addresses the limitations of existing tools by combining API-based mapping with an interactive interface. This innovative solution not only resolves synchronization issues but also enables efficient and accurate mapping of lunar data, paving the way for more advanced spatial analyses and visualization in lunar research.

4 Lunar Base Maps

Initially, we attempted to use QGIS for mapping and visualization of the lunar data. However, we faced challenges in converting the data into a dynamic 3D format suitable for interactive visualization and analysis. To overcome this limitation, we developed a prototype of our own software. The prototype software leverages latitude and longitude data obtained from NASA’s Moon API to generate dynamic 3D visualizations. Using a Celestron-based imaging system, the software extracts a 3D image of the lunar surface. Processed best-fit ratio data is then matched with the corresponding latitude and longitude to enable 3D elemental plotting. This approach not only resolved the challenges of dynamic visualization but also provided an effective way to integrate processed data with real lunar surface imagery for compositional analysis.



5 Challenges and Lessons Learned

Challenges

- Geotail contamination impacting data quality.
- Filtering and aggregation while maintaining spatial and temporal integrity.
- Initially, we attempted to use QGIS for mapping and visualization of the lunar data. However, we faced challenges in converting the data into a dynamic 3D format suitable for interactive visualization and analysis. This necessitated the development of a custom software solution to address these limitations.

Lessons Learned

- Rigorous preprocessing ensures reliable datasets.
- Aggregation techniques maintain a balance between spatial resolution and statistical robustness.
- Developing a custom software solution allowed us to dynamically visualize lunar data and enabled seamless integration of 3D surface imagery with processed elemental data.

6 Future Scope and Recommendations

- Automating preprocessing tasks like geotail removal and GTI filtering.
- Extending analysis to additional elemental ratios.
- Validate results with legacy datasets for improved accuracy.
- Develop advanced visualization techniques for better representation of 3D lunar maps.

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